IE2024 - Probability & Statistics

Descriptive Statistics

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MSc, MBA, PGDip, BSc (1st Class)



Descriptive Statistics

- Descriptive statistics summarize and describe the main features of a dataset quantitatively.
- They provide simple summaries about the sample and the measures.

Descriptive Statistics

- Here are some key concepts and measures used in descriptive statistics
  - Numerical Methods
    - Measures of Central Tendency
    - Measures of Variability (Dispersion)
    - Measures of Shape
  - Graphical Methods
    - Univariate Graphical Representation
    - Bivariate Graphical Representation

Numerical Methods

- Numerical methods in descriptive statistics involve summarizing and describing the essential features of a dataset. These methods provide simple summaries about the sample and the measures.
- Measures of Central Tendency
  - Mean
  - Median
  - Mode
  - Quartiles
- Measures of Variability (Dispersion)
  - Range
  - Variance & Standard Deviation
  - Interquartile Range (IQR)
- Measures of Shape
  - Skewness
  - Kurtosis

Mean

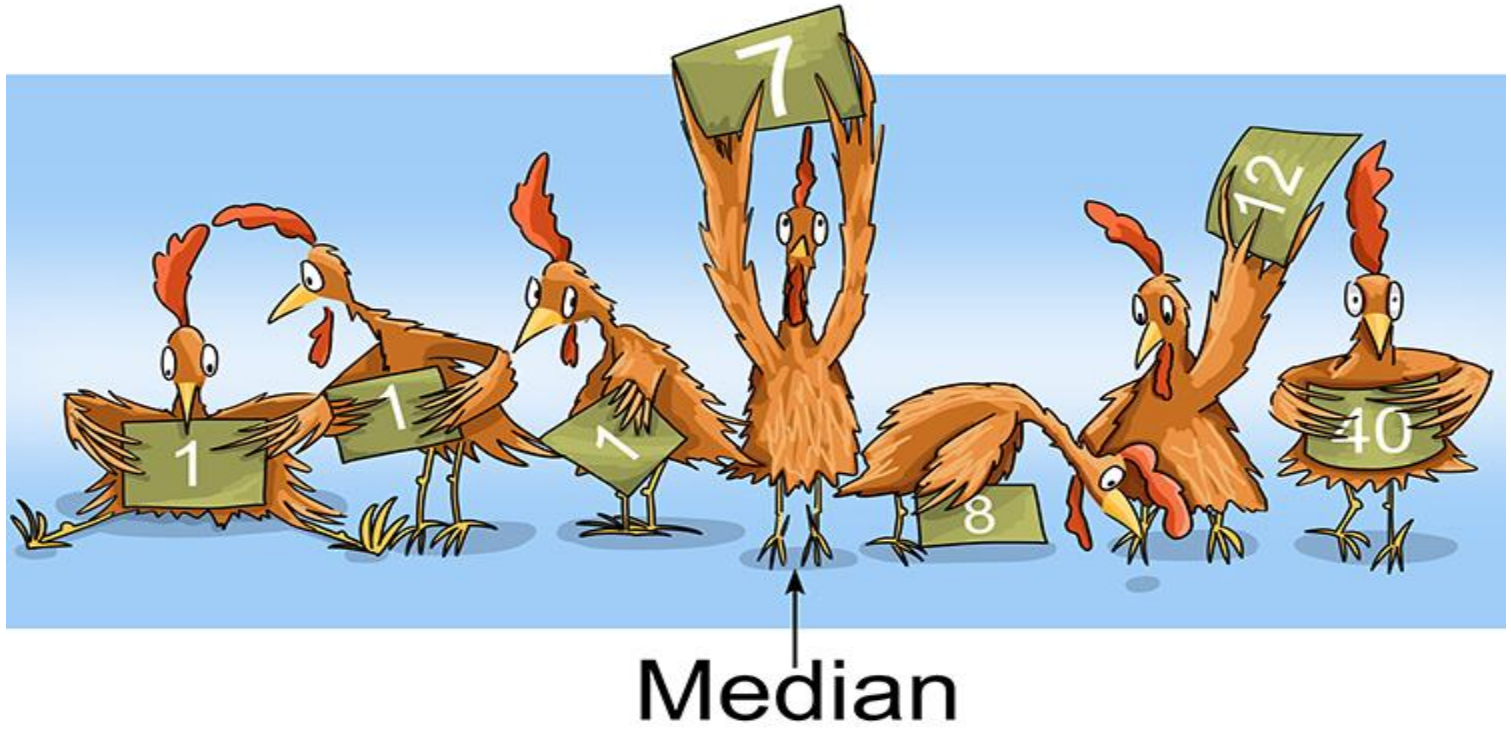
- Mean is also known as average of all the numbers in the data set which is calculated by below equation.



$$Pop\ Mean = \mu = \frac{\sum X_i}{N} \qquad Sample\ Mean = \bar{X} = \frac{\sum X_i}{n}$$

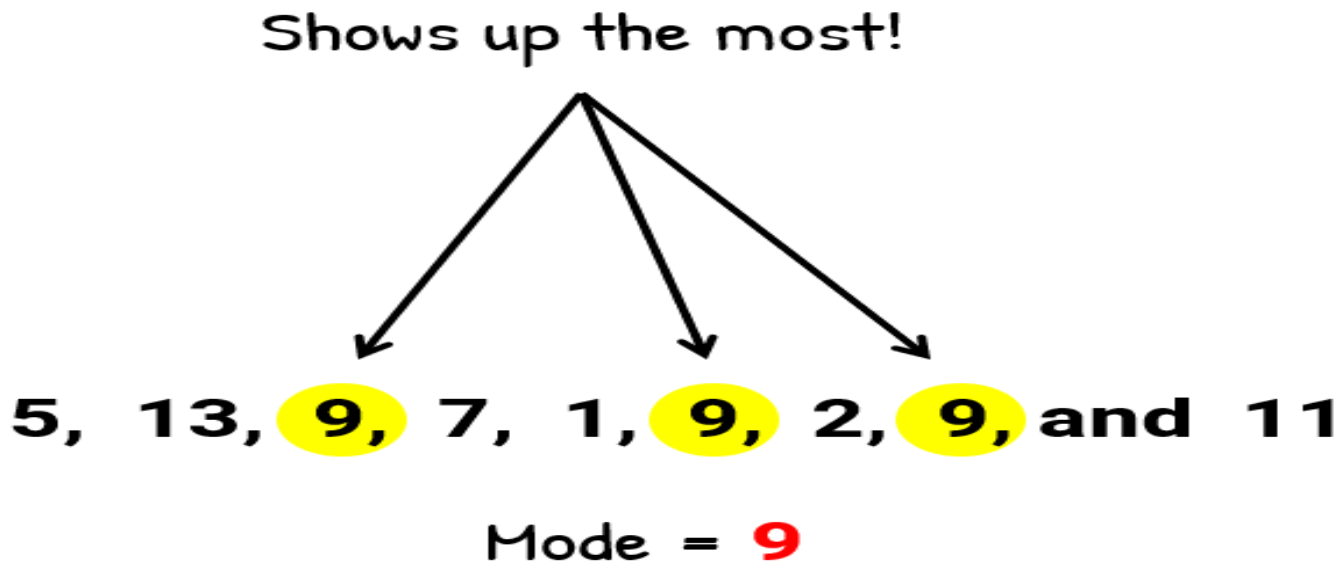
Median

- Median is mid value in an ordered data set.



Mode

- The mode is the value that has highest number of occurrences in a set of data.
- Unlike mean and median, mode can have both numeric and character data.



Quartiles

- Quartiles are specific percentiles that divide a dataset into four equal parts, each containing 25% of the data points.
- They are used to summarize the distribution of a dataset, providing insights into its spread and central tendency.
- Quartiles are values that divide a dataset into four equal parts. The three main quartiles are:
  - First Quartile (Q1): Below which 25% of the data points lie.
  - Second Quartile (Q2 or Median): Below which 50% of the data points lie.
  - Third Quartile (Q3): Below which 75% of the data points lie.

Quartiles

- To calculate quartiles, follow these steps:
  - Arrange the Data: Sort the dataset in ascending order.
  - Find the Median (Q2):
    - If the number of data points (N) is odd, the median is the middle value.
    - If N is even, the median is the average of the two middle values.
  - Find Q1 and Q3:
    - Q1: The median of the lower half of the dataset (excluding Q2 if N is odd).
    - Q3: The median of the upper half of the dataset (excluding Q2 if N is odd).

Range

- The range is a simple measure of dispersion that indicates the spread of a dataset by calculating the difference between the maximum and minimum values.
- It provides a basic understanding of how spread out the data points are.

$$Range = Maximum\ value - Minimum\ value$$

Interquartile Range (IQR)

- Interquartile Range is the difference between Q3 and Q1.

$$IQR = Q_3 - Q_1$$

Interquartile Range (IQR)

- Expected minimum for a data range
$$Q_1 - 1.5 \times IQR$$
- Expected maximum for a data range
$$Q_3 + 1.5 \times IQR$$
- Outside this range, the values are considered as significant outliers.

Variance & Standard Deviation

- Variance is the numerical values that describe the variability of the observations from its arithmetic mean.
- Standard Deviation is the square root of the variance.

$$V(X) = \frac{1}{N}\sum_{i=1}^{N}(X_i - \mu)^2 \qquad STD(X) = \sqrt{V(X)}$$

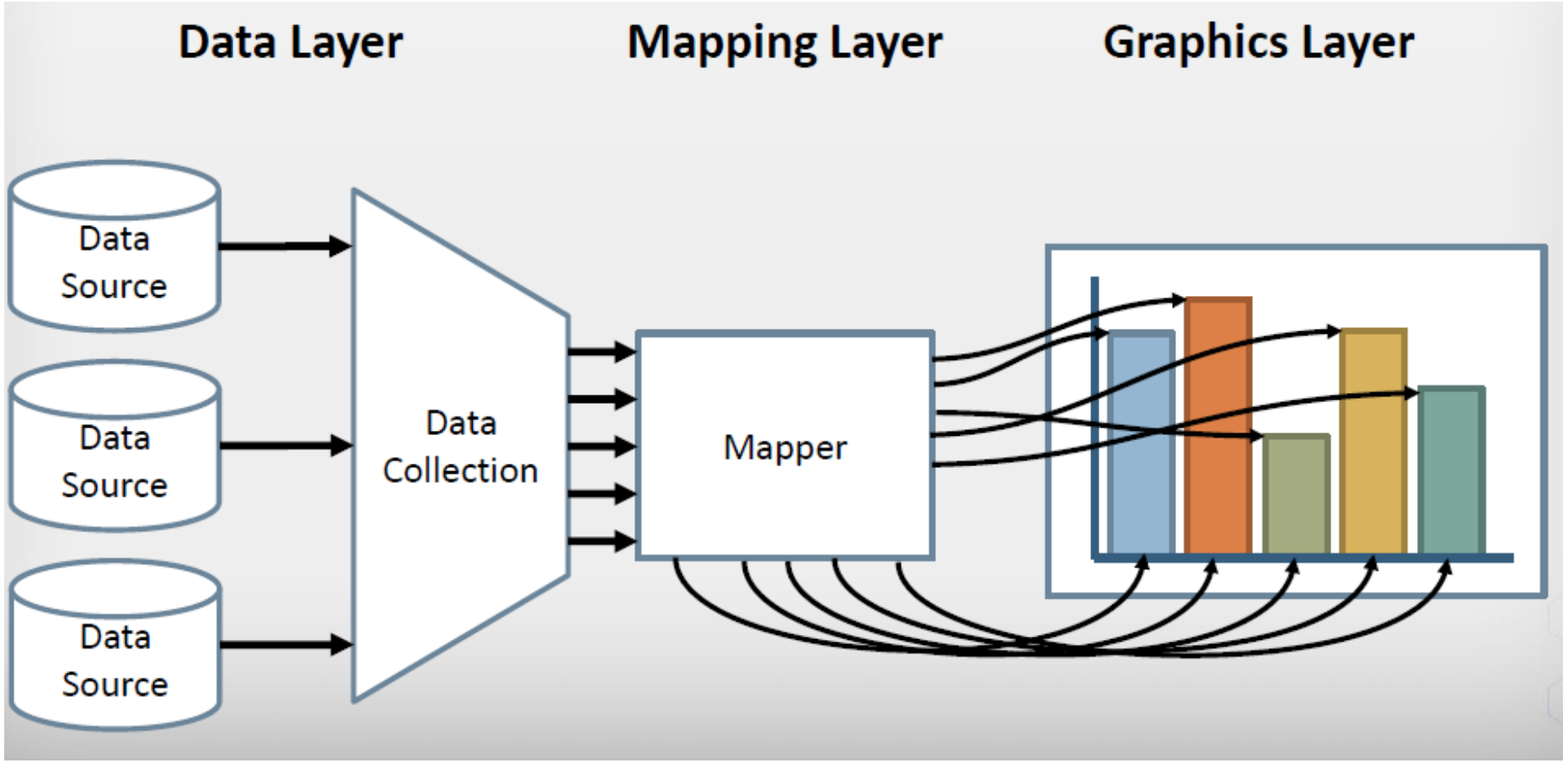
Variance & Standard Deviation

- Sample variance

$$V(X) = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

Data Visualization

- Process of converting data into a graphical representation.



Data Visualization

- Data visualization gives us a clear idea of what the information means by giving it visual context through maps or graphs.
- This makes the data more natural for the human mind to comprehend and therefore makes it easier to identify trends, patterns, and outliers within large data sets.

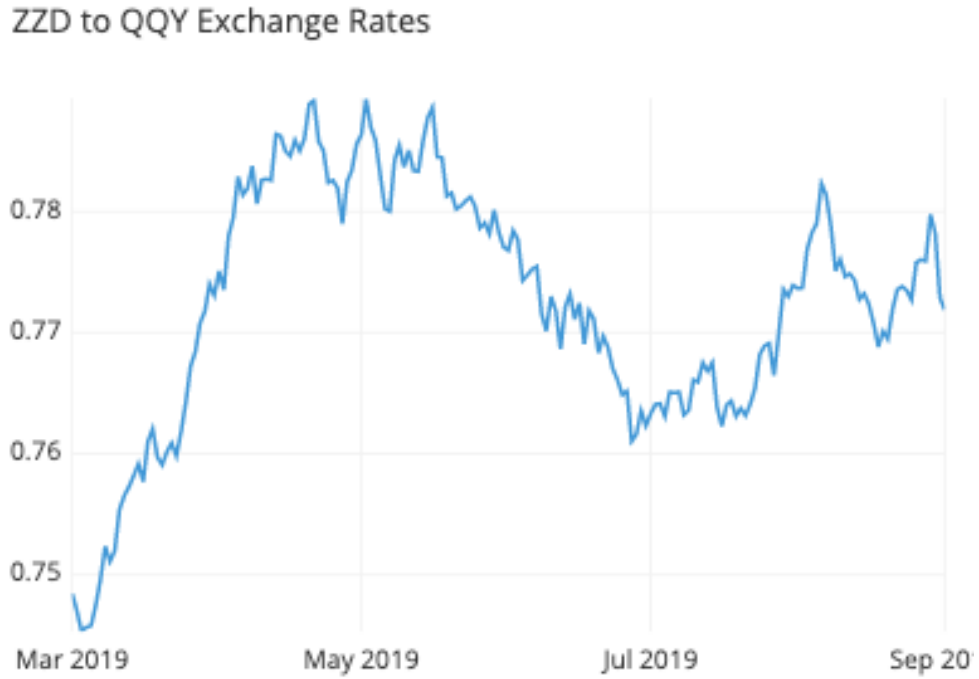


Data Visualization

- Type of variables is one of the main factors to be considered when visualizing
  - Numerical variables (Ex- Age, Height, Weight)
  - Categorical variables (Ex- Gender, Color, District, Country)

Line Chart

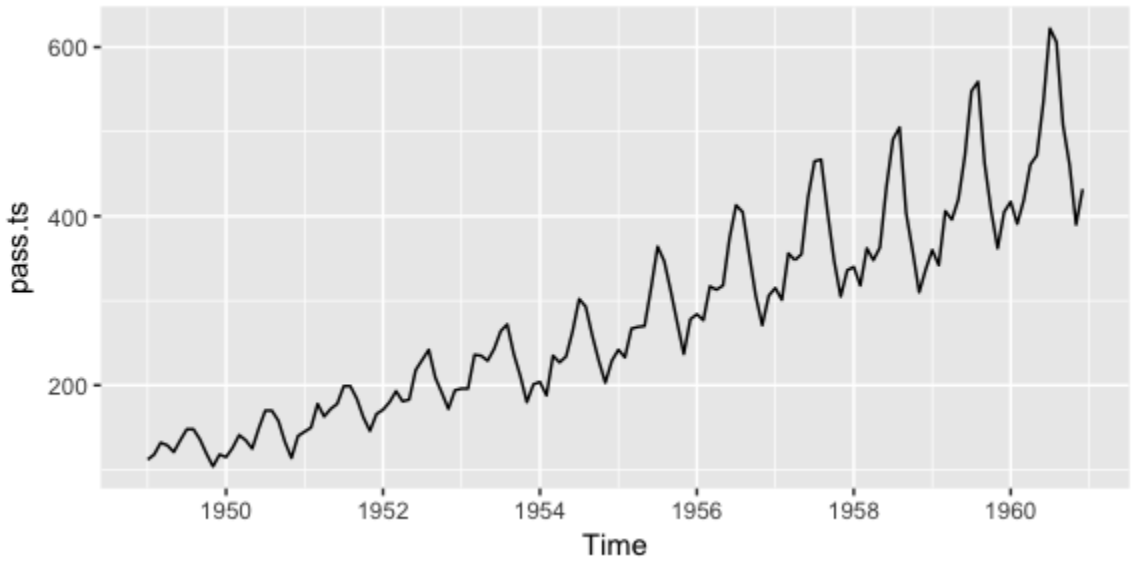
- A line chart or line plot or line graph or curve chart is a type of chart which displays information as a series of data points called 'markers' connected by straight line segments.



Line Chart

- Important patterns which can be observed

Seasonality



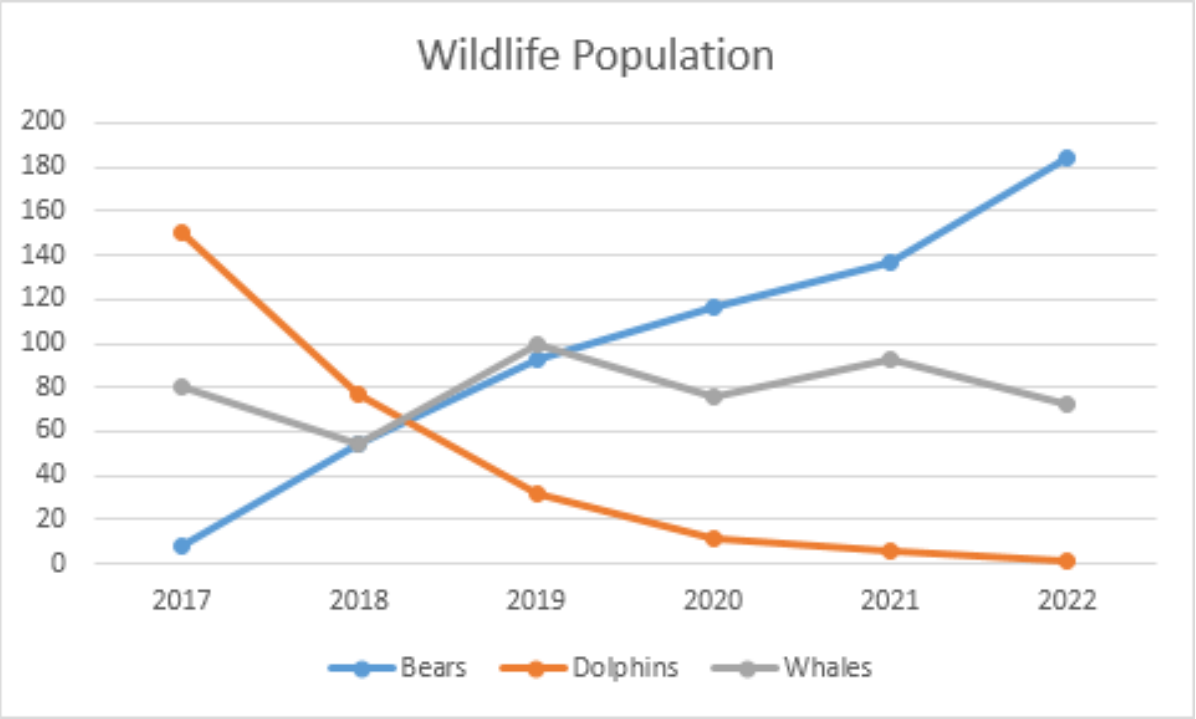






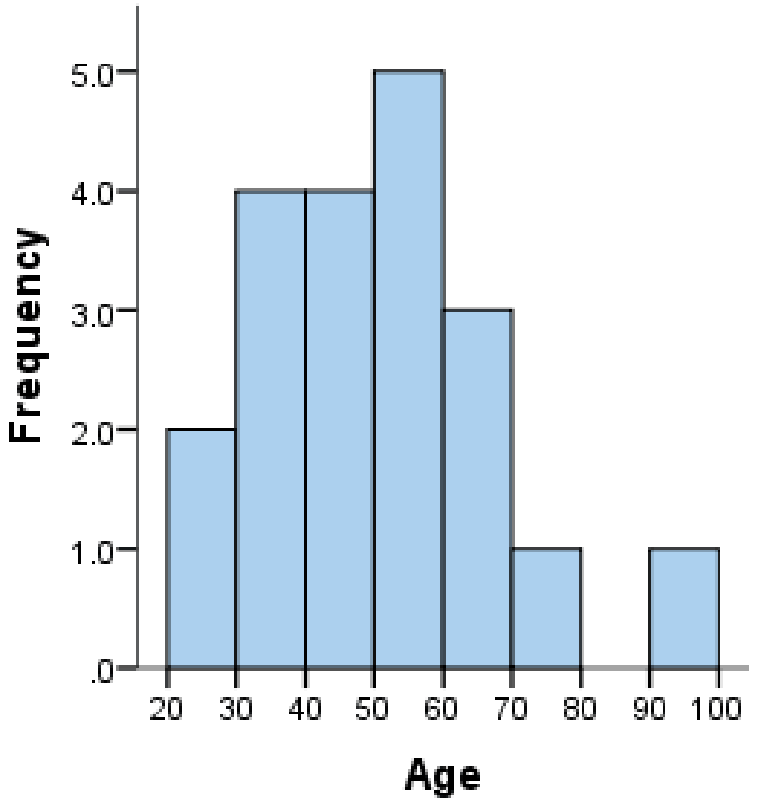
Line Chart

- Line charts are important when comparing several charts.



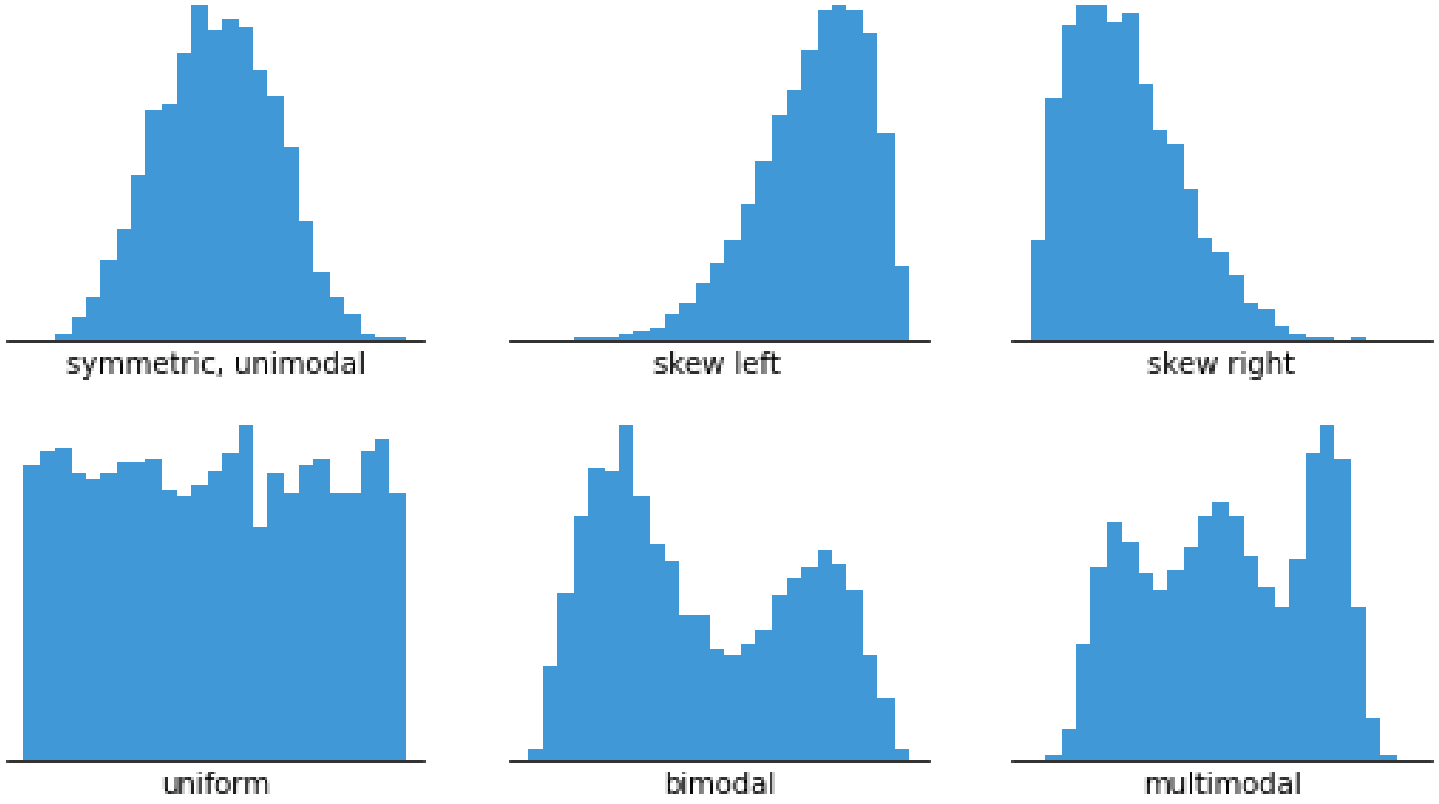
Histogram

- A histogram is an approximate representation of the distribution of numerical data.

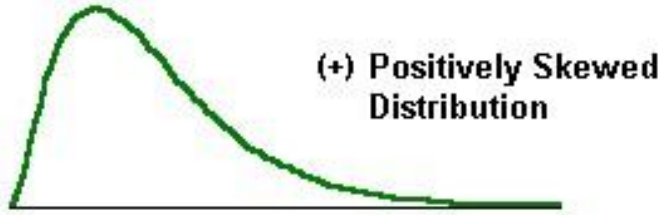


Histogram

- Different shapes can be observed when we draw histograms.



Skewness and Kurtosis



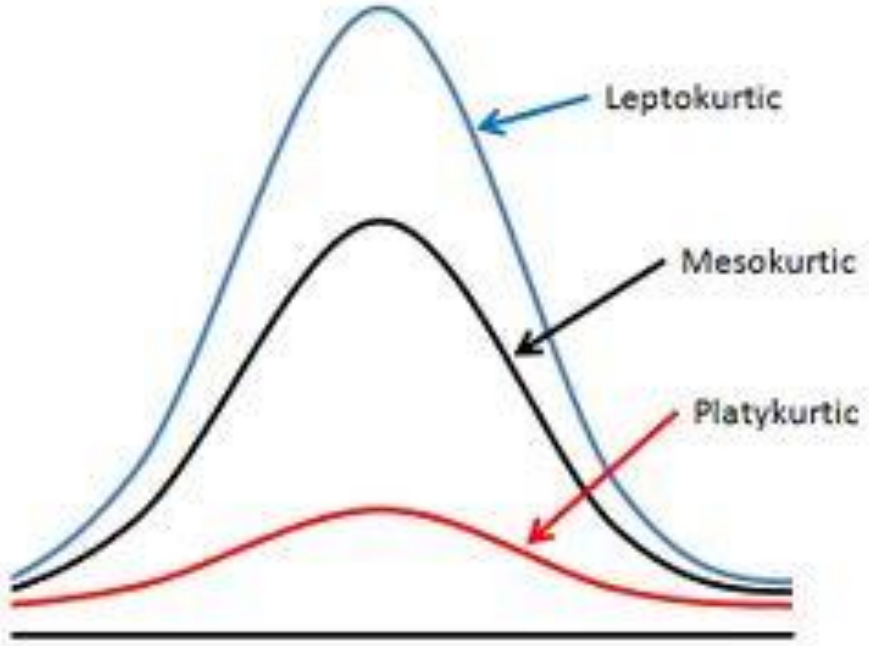
- Two measurements are Skew and Kurtosis



$$\text{Skew} = \frac{n}{(n-1)(n-2)}\sum\left(\frac{x_j - \bar{x}}{s}\right)^3$$

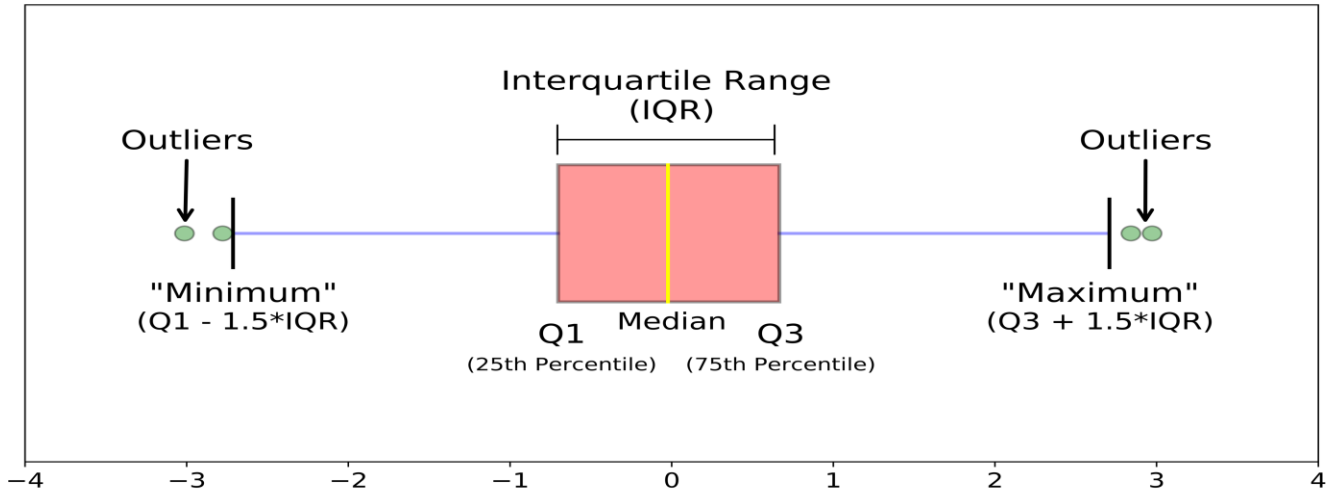
$$\text{Kurtosis} = \left\{\frac{n(n+1)}{(n-1)(n-2)(n-3)}\sum\left(\frac{x_j - \bar{x}}{s}\right)^4\right\} - \frac{3(n-1)^2}{(n-2)(n-3)}$$

- **Mesokurtic** distributions have excess kurtosis of zero.
- **Platykurtic** distributions (light tails) have negative excess kurtosis.
- **Leptokurtic** distributions (heavy tails) have positive excess kurtosis.



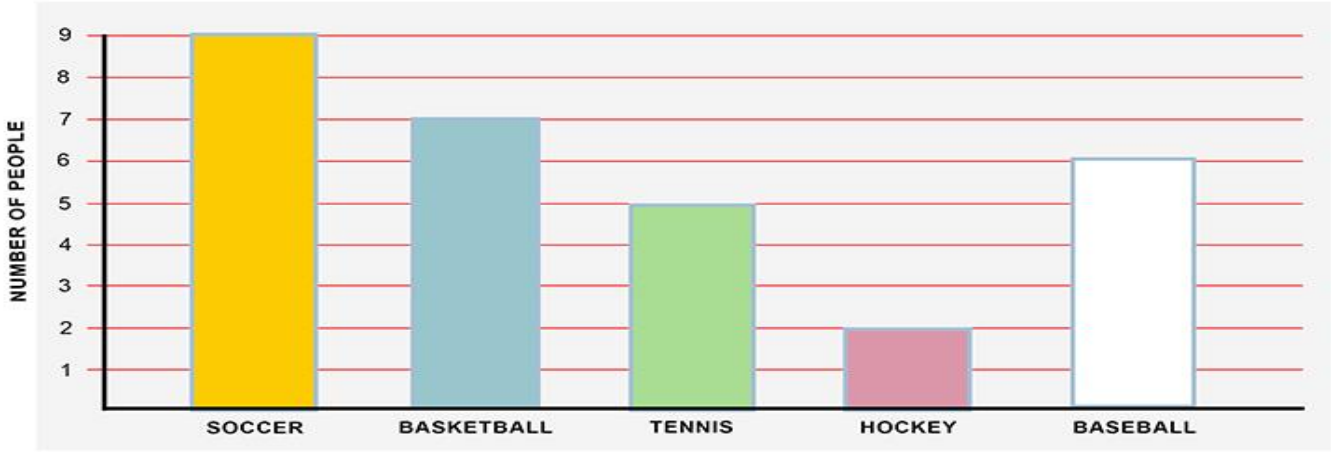
Boxplot

- In descriptive statistics, a box plot or boxplot is a method for graphically depicting groups of numerical data through their quartiles (5 Number Summary).
- They are showing outliers as well.



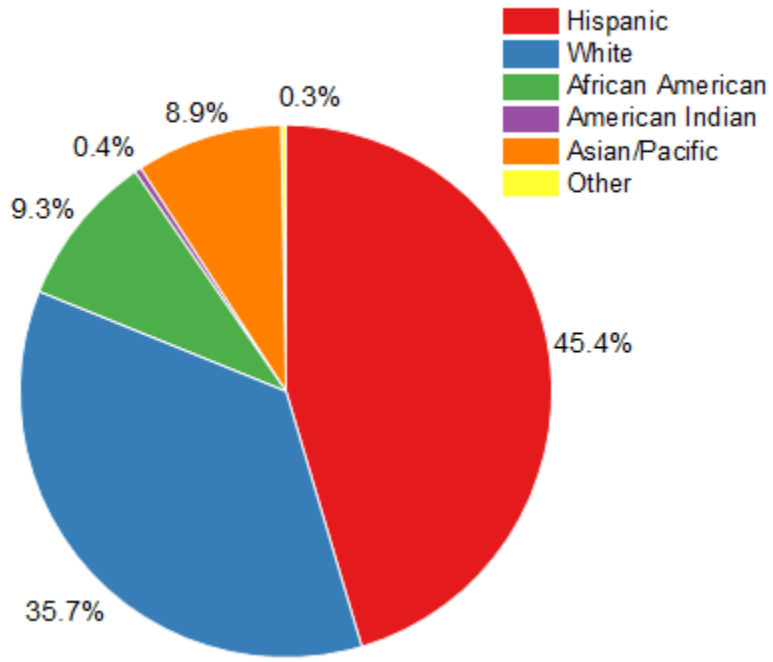
Bar Chart

- A bar chart or bar graph is a chart or graph that presents categorical data with rectangular bars with heights or lengths proportional to the values that they represent.
- The bars can be plotted vertically or horizontally.



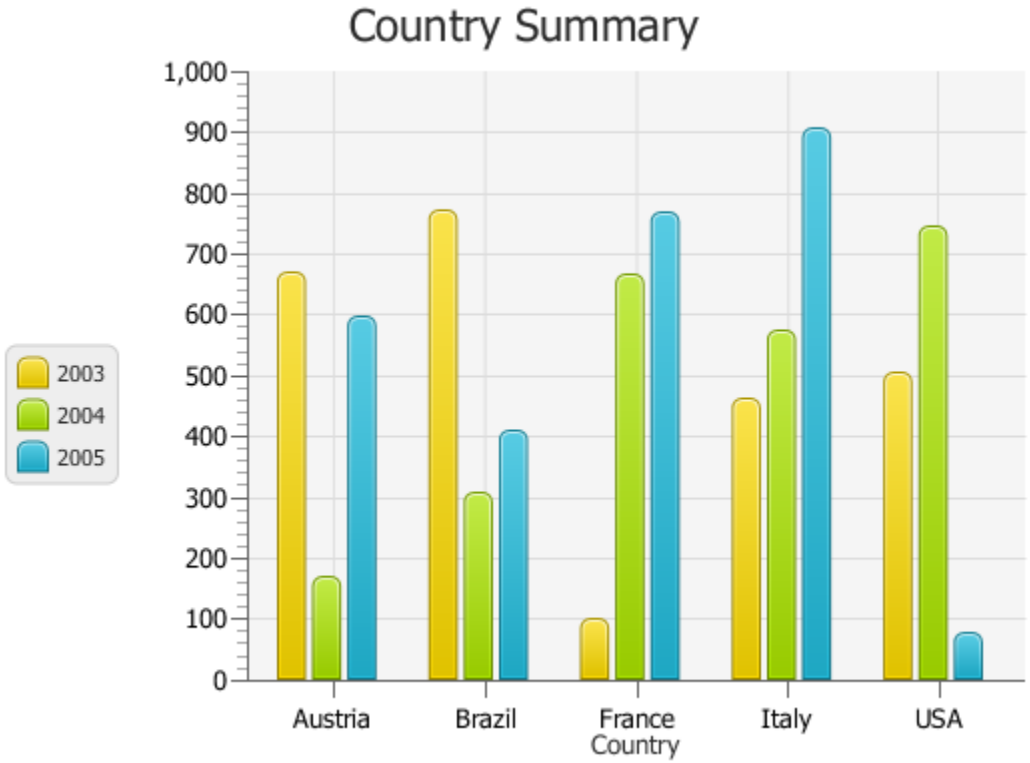
Pie Chart

- A pie chart is a circular statistical graphic, which is divided into slices to illustrate numerical proportion.



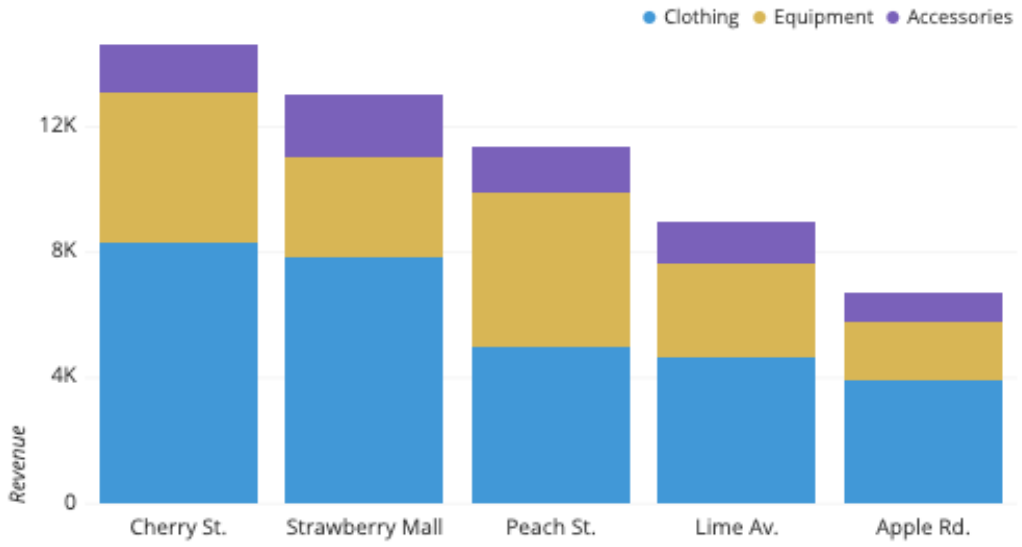
Multiple Bar Chart

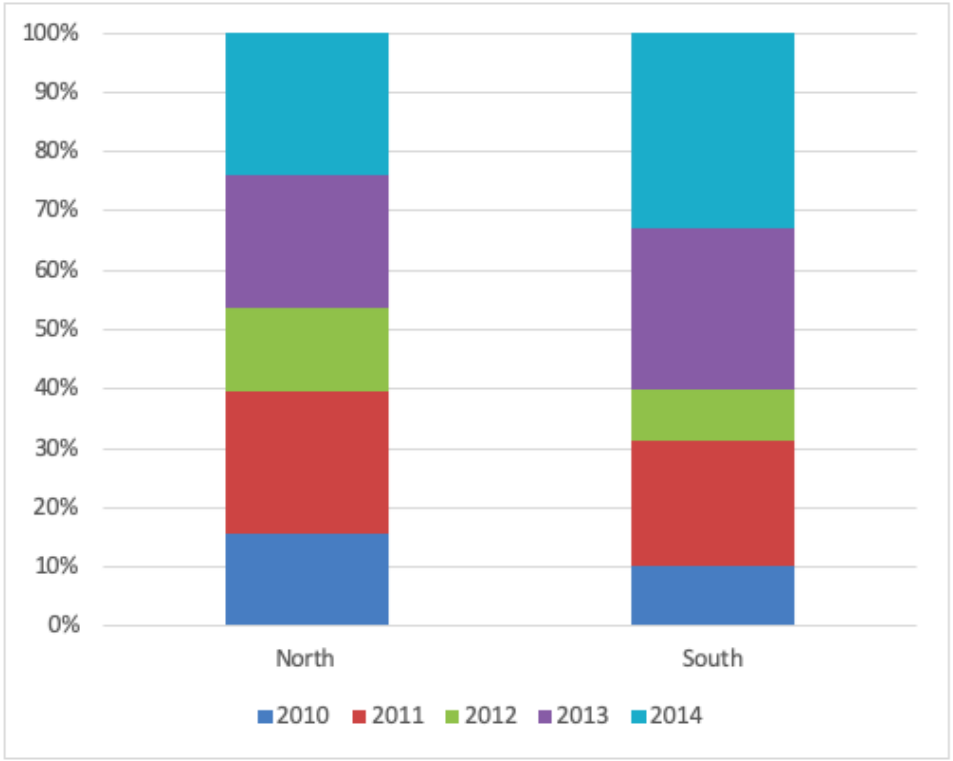
- Side by side multiple bar charts show how two or more categories are related



Stacked Bar Chart

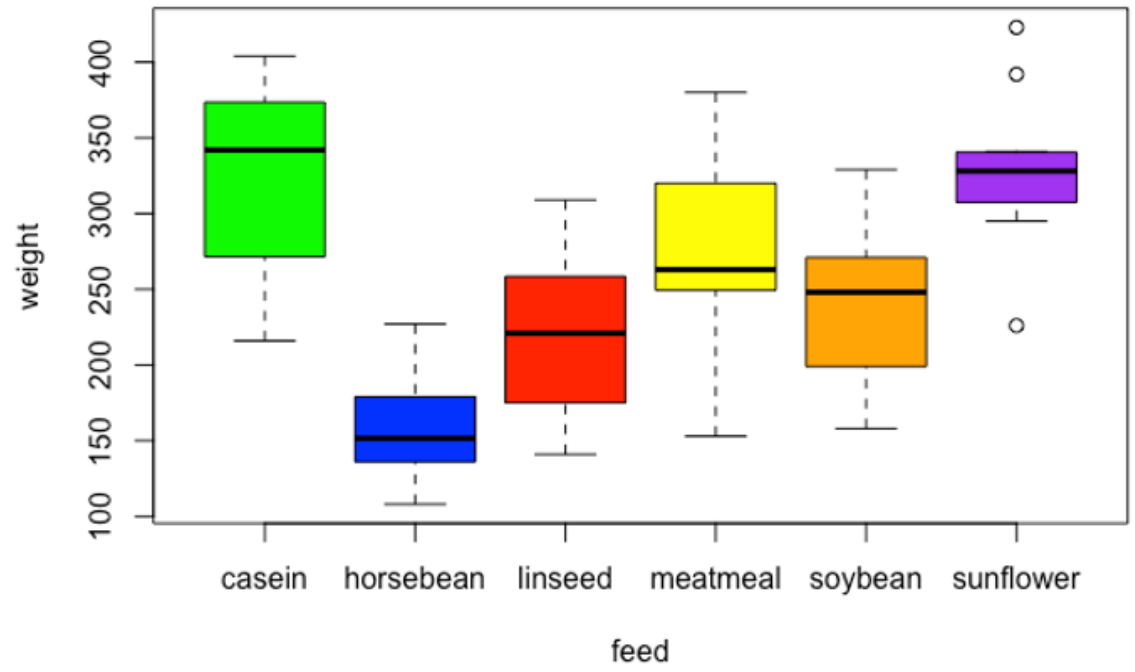
- In a stacked bar chart, parts of the data are adjacent or stacked. Each bar displays a total amount, broken down into sub-amounts.





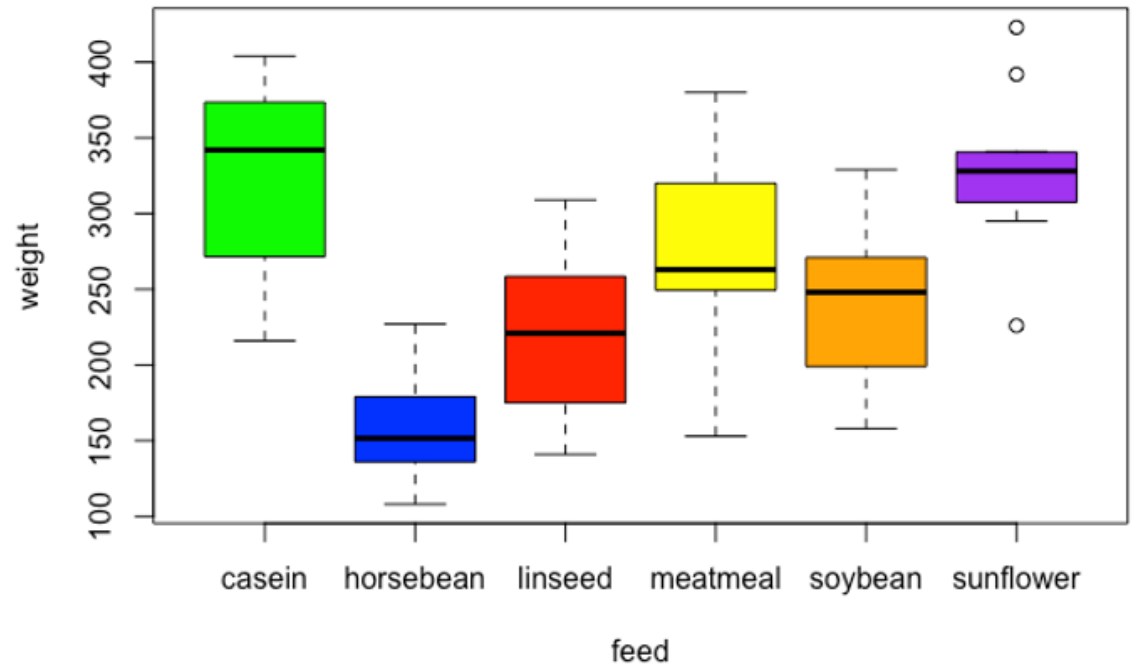
Side by Side Boxplots

- Side-By-Side boxplots are used to display the distribution of several quantitative variables or a single quantitative variable along with a categorical variable.



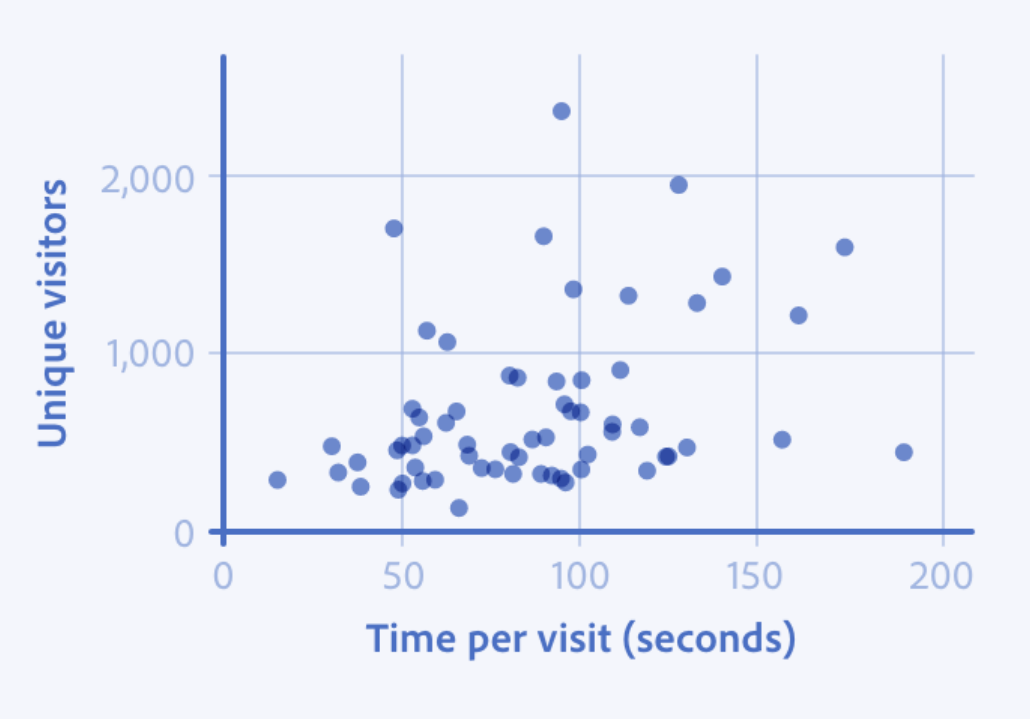
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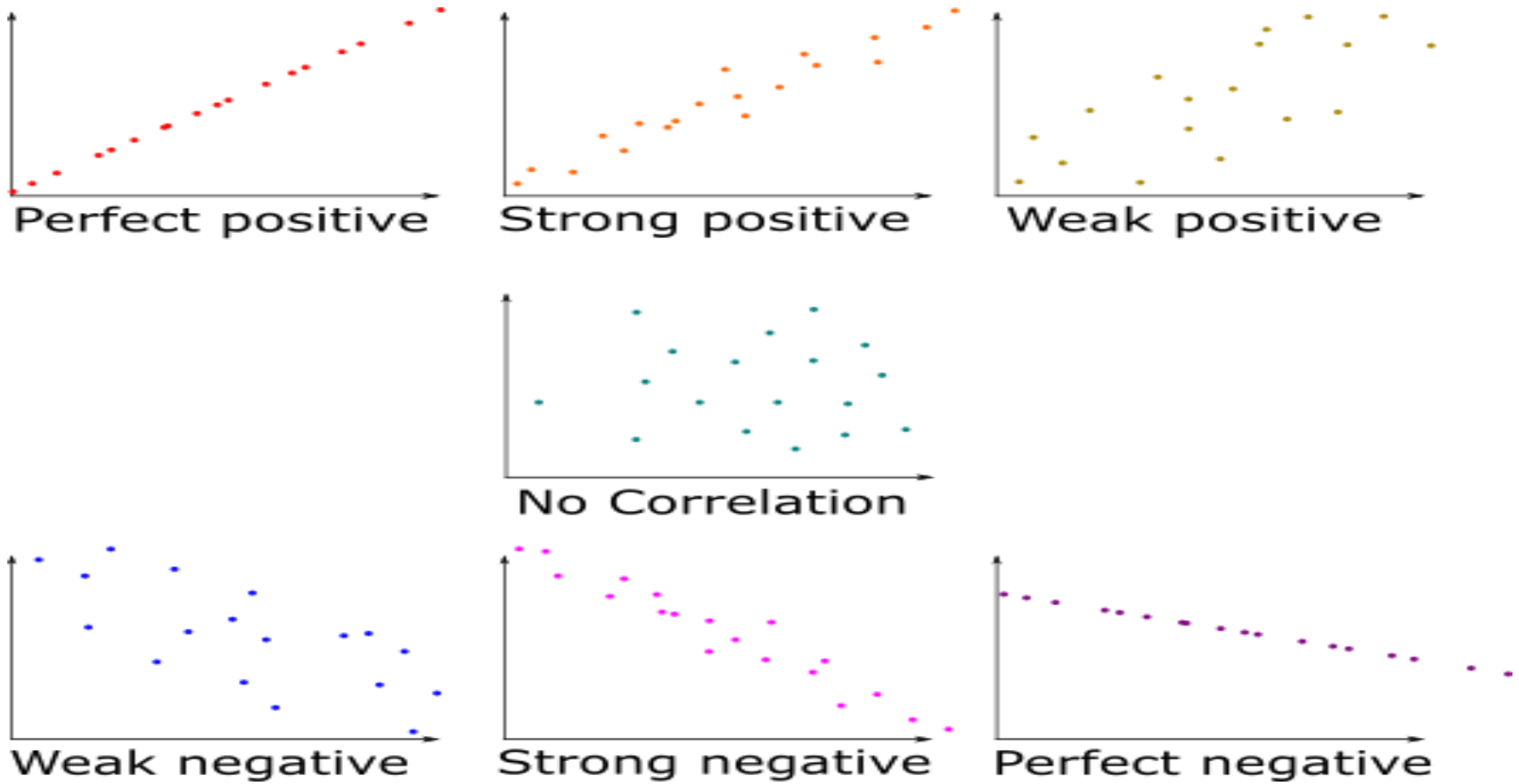


Scatter Plots

- A scatter plot is a type of plot or mathematical diagram using Cartesian coordinates to display values for typically two variables for a set of data.



Scatter Plots



Covariance of Two Numerical Variables

- Covariance will measure joint variation of two numerical variables. This is a measure of the relationship between two variables.

$$COV(X,Y) = \frac{1}{N}\sum_{i=1}^{N}(X_i - \mu_X)\,(Y_i - \mu_Y)$$

Correlation of Two Numerical Variables

- Correlation is also a measurement of the relationship between two numerical variables.
- It is lying between -1 and +1.

$$CORR(X,Y) = \frac{COV(X,Y)}{\sqrt{V(X)V(Y)}}$$

Correlation of Two Numerical Variables

- Correlation coefficients are used to measure how strong a relationship is between two variables.

